

Chemical Identity Lost in Regulation

A Study of Semantic Interoperability in European Chemical Substance Data

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The problem

In chemistry, substance identity is exact: structure-based identifiers (**InChI** / **InChIKey**) link a molecule unambiguously across PubChem, ChEBI and beyond.

In EU regulation (**REACH**, **CLP**), a substance may be a single molecule, a mixture or **UVCB**, a **group** (e.g. PFAS, with no explicit set membership), or an analytical sum parameter — identified only by **CAS numbers** and textual names that carry no structure.

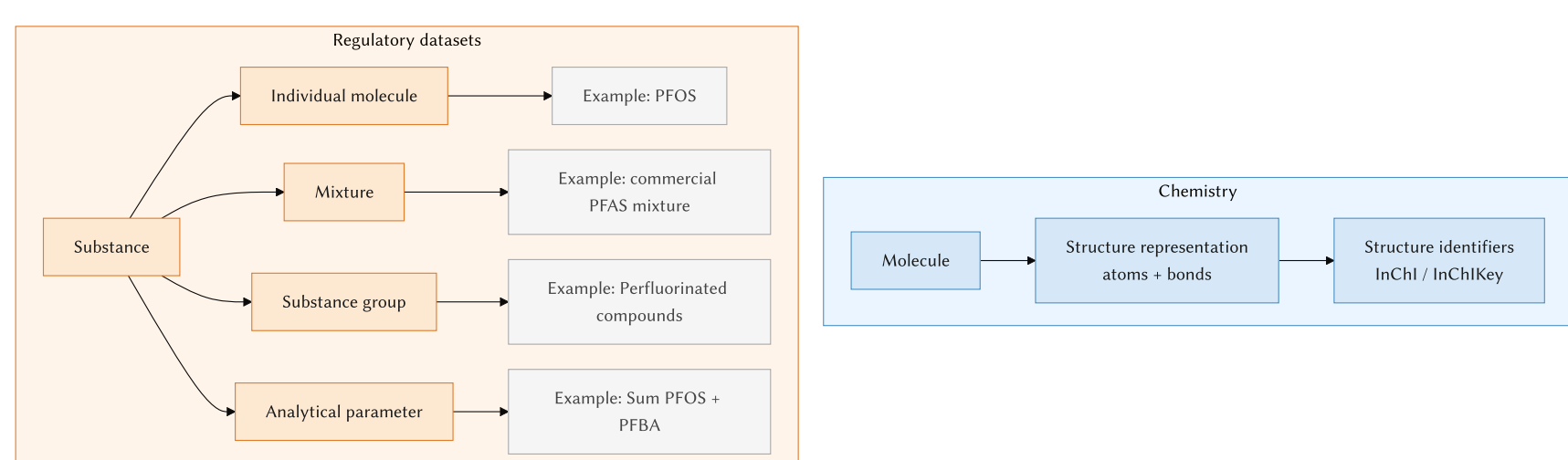


Fig. 1 — Structure-based chemical identity vs. heterogeneous regulatory substance types.

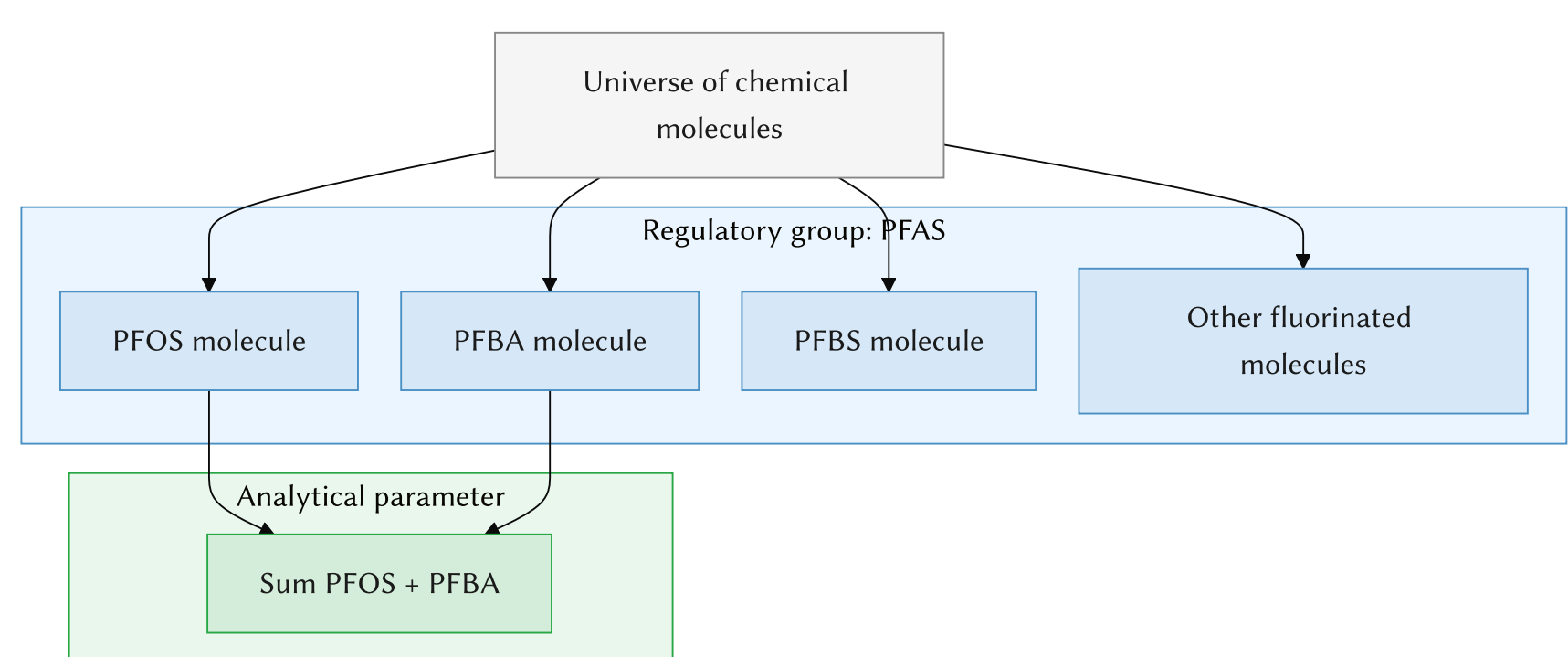


Fig. 2 — PFAS as a regulatory group without set-membership semantics; Sum PFOS + PFBA a derived analytical parameter.

Approach

We integrated **18 ECHA-based datasets** (41,813 records): obligation and activity lists, REACH registrations, the EU Pesticides Database, OSPAR and a Flemish administrative list. Each entry was linked to InChIKey via PubChem and assigned to a **seven-tier linkability taxonomy**.

Substance names were matched to ~4,800 ChemOnt classes using four sentence-transformer models and Claude Sonnet 4.6, benchmarked against 28,204 ClassyFire assignments.

62.5%

link to a defined structure (tier 1)

37.5%

non-structure-based entries

39.2%

best LLM Hit@1 (still a minority)

Linkability (RQ1)

Of 33,452 unique entries, **62.5% (20,909)** link directly via InChIKey. The other **37.5%** resist structure-based linking: 11.6% unresolved CAS (tier 4), 10.8% mixtures or UVCBs (tier 5), and 14.9% lack any usable identifier (tier 7).

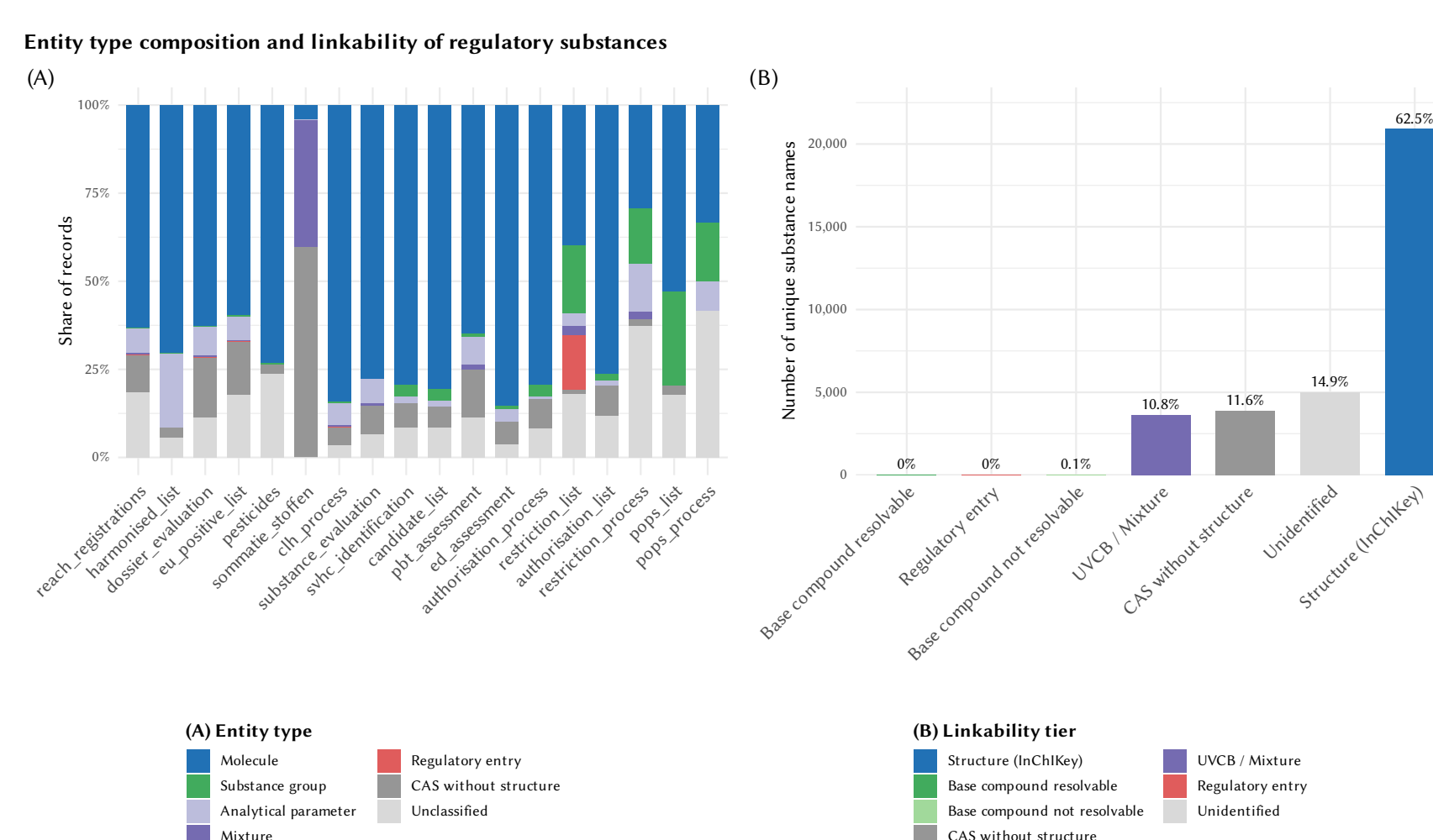


Fig. 3 — Entity-type composition (A) and seven-tier linkability (B) across 18 sources.

Semantic methods fail (RQ3)

Embedding-based matching collapses: the best model reaches **Hit@1 = 5.75%**; all others score below 1%. Claude Sonnet 4.6 reaches **39.2%** — a seven-fold gain, but it still misclassifies the majority. Text cannot substitute for structure.

Model	Hit@1	Hit@3	Hit@5	Hit@10
all-MiniLM-L6-v2	0.0575	0.0986	0.1234	0.1621
all-mpnet-base-v2	0.0093	0.0152	0.0184	0.0247
SciBERT	0.0040	0.0071	0.0102	0.0131
BioBERT	0.0047	0.0077	0.0099	0.0150
Claude Sonnet 4.6	0.392	—	—	—

Table 1 — ChemOnt classification (direct-parent level). Only Hit@1 is comparable across rows.

What works: the knowledge graph (C4, C5)

~17,500 regulatory substances (SKOS concepts) are integrated with structure-based identity, ChemOnt classification and ChEBI biological roles — **~46,600 cross-domain links**, enabling composite priority scoring that is infeasible on fragmented data.

LLM-based scope mapping yields **304 validated SKOS triples** (285 broadMatch, 18 exactMatch, 1 narrowMatch) as a legislative–ChemOnt cross-walk.

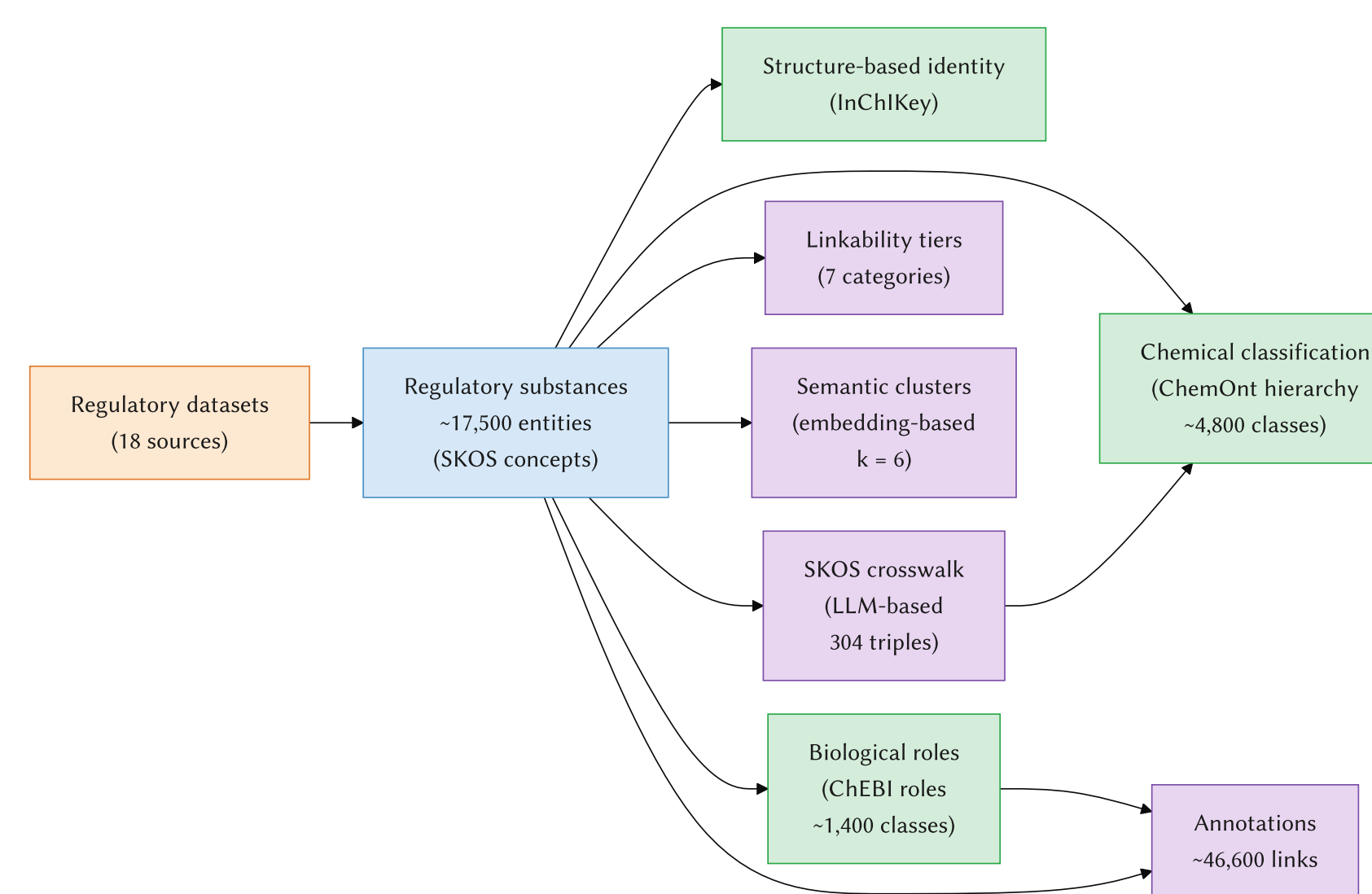


Fig. 4 — Integrated knowledge graph linking identity, classification and biological roles.

Research questions

RQ1 To what extent can regulatory substances be linked to a unique chemical structure?

RQ2 How reliable are CAS numbers as cross-dataset identifiers?

RQ3 Can embedding-based or LLM methods compensate for absent structure identifiers?

CAS unreliability (RQ2)

70.0% of structure-resolved InChIKeys appear in only one source — CAS is insufficient as a shared key.

Ambiguity compounds this: 195 InChIKeys map to multiple CAS numbers (up to seven); 4 CAS numbers resolve to two distinct InChIKeys (e.g. toluene diisocyanate → two positional isomers).

CAS numbers introduce ambiguity rather than resolving it.

Conclusions & recommendations

Over one-third of entries cannot be linked to a structure; CAS adds ambiguity; neither embeddings nor LLMs can replace structural information. Manual reconciliation (~3,150 person-years for REACH) is infeasible. **Harmonisation must happen at the source, not downstream.**

1 Require structure-based identifiers (InChIKey; MInChI for mixtures) across regulatory instruments.

2 Mandate explicit substance-group representation using formal ontologies such as ChemOnt.

3 Publish regulatory data as linked data with persistent URIs and SKOS links (FAIR / Interoperable Europe Act).

Code & data: github.com/gezever/A-Substance-Is-Not-Always-a-Substance

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AI declaration: Claude Sonnet 4.6 was used as the object of study for the LLM experiments (~\$100 API) and as a development tool. All content was reviewed by the authors. CC BY 4.0.



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